

4.5 Browser IndexedDB Architecture & Database Schema (PrismDB)

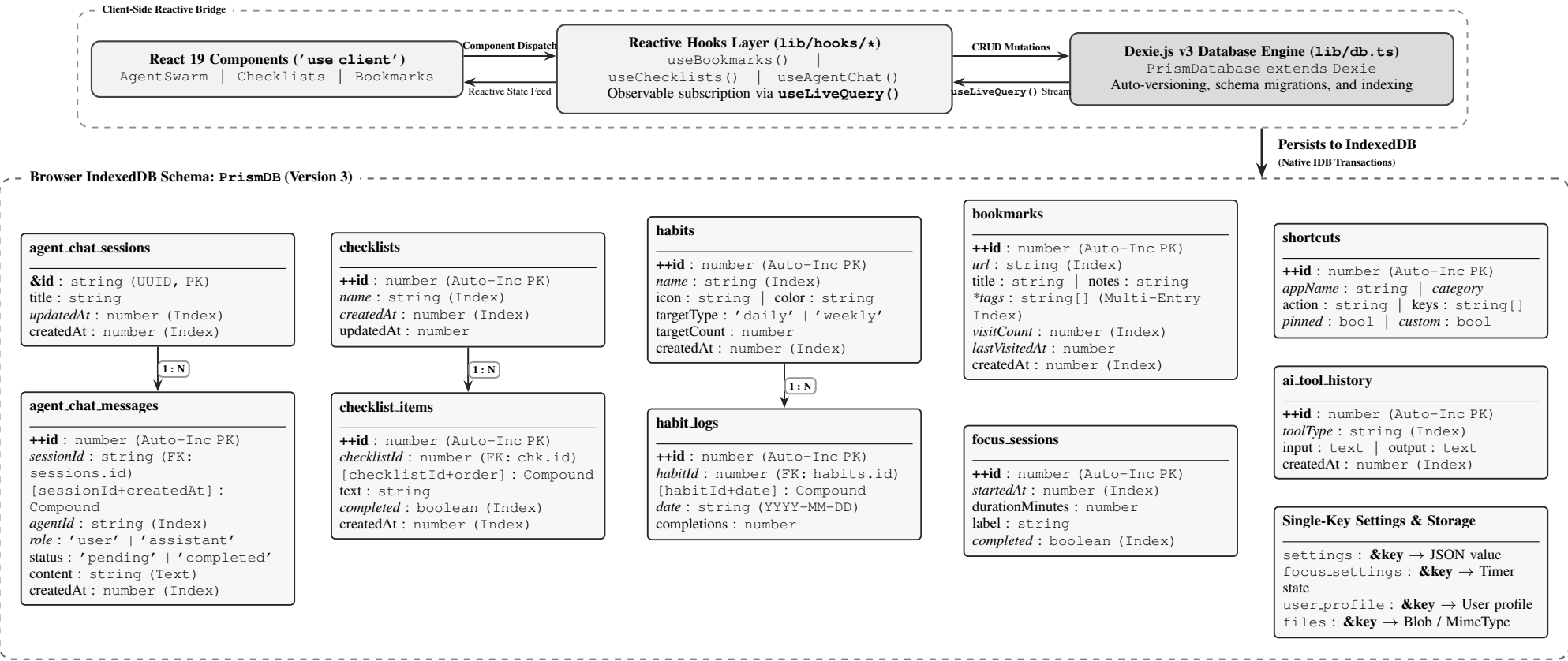


Figure 4.4: Browser IndexedDB Database Schema and Reactive Data Architecture (PrismDB)

Database Schema & Reactive Engine Glossary

PrismDB: Local client-side database built on IndexedDB storing sessions, habits, tasks, and settings securely in-browser.

IndexedDB (IDB): W3C standard browser NoSQL object store providing high-capacity transactional persistence.

Dexie.js v3: Minimalist TypeScript wrapper over IndexedDB handling migrations, indices, and transactions.

PK (++id / &id): Primary Key; ++id auto-increments integers; &id designates unique string/UUID keys.

Idx / Compound Index: Secondary B-tree index; [a+b] enables high-speed compound filtering without full scans.

Multi-Entry (*tags): Array index where each element is indexed individually for rapid membership search.

useLiveQuery(): Reactive hook subscribing React components to live query results, auto-re-rendering on data change.

FK (Foreign Key): Logical relational link associating child entity records with parent table primary keys (1 : N).

CRUD Transactions: Atomic Create, Read, Update, and Delete operations executed with ACID guarantees in browser.